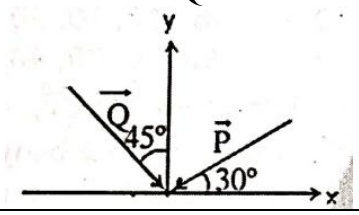
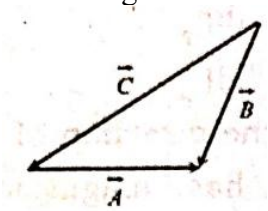
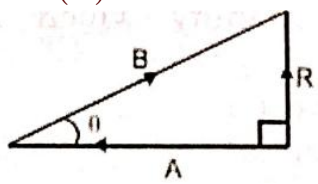
Question	Which of the following is a scalar quantity?
Type	multiple choice
Option	Displacement
Option	Electric field
Option	Acceleration
Option	Work
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A physical quantity which has a direction:
Type	multiple choice
Option	must be a vector
Option	may be a vector
Option	must be a scalar
Option	none of the above
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following physical quantities is an axial vector?
Type	multiple choice
Option	Displacement
Option	Force
Option	Velocity
Option	torque
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The direction of the angular velocity vector is along:
Type	multiple choice
Option	Along the tangent of circular path
Option	Along the direction of radius vector
Option	Opposite to the direction of radius vector
Option	Along the axis of rotation
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Vector P and Q are shown in the figure. Find the angle between them 
Type	multiple choice
Option	105°
Option	75°
Option	110°
Option	120°
Answer	1
Solution	

Positive Marks	4
Negative Marks	1
Question	If $\hat{n}$ is a unit vector in the direction of the vector $\vec{A}$, then:
Type	multiple choice
Option	$\hat{n} = \frac{\vec{A}}{\lvert\vec{A}\rvert}$
Option	$\hat{n} = \vec{A}\lvert\vec{A}\rvert$
Option	$\hat{n} = \frac{\lvert\vec{A}\rvert}{\vec{A}}$
Option	None of the above
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	If a unit vector is represented by $0.5\hat{i} - 0.8\hat{j} + C\hat{k}$ then the value of C is
Type	multiple choice
Option	$\sqrt{0.01}$
Option	$\sqrt{0.11}$
Option	1
Option	$\sqrt{0.39}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	If a particle moves from point P (2,3,5) to point Q (3,4,5). Its displacement vector is
Type	multiple choice
Option	$\hat{i} + \hat{j} + 10\hat{k}$
Option	$\hat{i} + \hat{j} + 5\hat{k}$
Option	$\hat{i} + \hat{j}$
Option	$2\hat{i} + 4\hat{j} + 6\hat{k}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The value of λ for which the vectors $3\hat{i} - 6\hat{j} + \hat{k}$ and $2\hat{i} - 4\hat{j} + \lambda\hat{k}$ are parallel is
Type	multiple choice
Option	$\frac{2}{3}$
Option	$\frac{3}{2}$
Option	$\frac{5}{2}$

Option	$\frac{2}{5}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Find a vector in the direction of vector $\vec{a} = 3\hat{i} - 4\hat{j}$ that has magnitude 10 units
Type	multiple choice
Option	$6\hat{i} + 8\hat{j}$
Option	$6\hat{i} - 8\hat{j}$
Option	$3\hat{i} - 4\hat{j}$
Option	$3\hat{i} + 4\hat{j}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The vector sum of two force F_1 and F_2 is minimum when the angle θ between their positive directions is
Type	multiple choice
Option	$\frac{\pi}{4}$
Option	$\frac{\pi}{3}$
Option	$\frac{\pi}{2}$
Option	π
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	$\vec{A} + \vec{B} = \vec{R}$ and $\vec{A} - \vec{B} = \vec{D}$ then find angle between $\vec{A}$ and $\vec{B}$ if R = D
Type	multiple choice
Option	$\frac{\pi}{2}$
Option	$\frac{\pi}{3}$
Option	$\frac{\pi}{4}$
Option	π
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	$\vec{A} = 2\hat{i} + \hat{j},\ B = 3\hat{j} - \hat{k}$ and $\vec{C} = 6i - 2\hat{k}$. Value of $\vec{A} - 2\vec{B} + 3\vec{C}$ would be
Type	multiple choice
Option	$20\hat{i} + 5\hat{j} + 4\hat{k}$

Option	$20\hat{i}-5\hat{j}-4\hat{k}$
Option	$4\hat{i}+5\hat{j}+20\hat{k}$
Option	$5\hat{i}+4\hat{j}+10\hat{k}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The vector sum of two vectors of magnitude 10 N and 6 N can be
Type	multiple choice
Option	2 N
Option	8 N
Option	18 N
Option	20 N
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The sum and difference of two perpendicular vectors of equal lengths are
Type	multiple choice
Option	of equal lengths and have an acute angle between them
Option	of equal length and have an obtuse angle between them
Option	also perpendicular to each other and are of different lengths
Option	also perpendicular to each other and are of equal lengths.
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	For the figure 
Type	multiple choice
Option	$\vec{A}+\vec{B}=\vec{C}$
Option	$\vec{B}+\vec{C}=\vec{A}$
Option	$\vec{C}+\vec{A}=\vec{B}$
Option	$\vec{A}+\vec{B}+\vec{C}=0$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A truck travelling due north at 20 m/s turns west and travels at the same speed. The change in its velocity be
Type	multiple choice
Option	$40\,\mathrm{m/s}\ \mathrm{N-W}$
Option	$20\sqrt{2}\ \mathrm{m/s}\ \mathrm{N-W}$
Option	$40\ \mathrm{m/s}\ \mathrm{S-W}$
Option	$20\sqrt{2}\ \mathrm{m/s}\ \mathrm{S-W}$

Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	If the sum of two-unit vectors is a unit vector, then magnitude of its difference is
Type	multiple choice
Option	$\sqrt{2}$
Option	$\sqrt{3}$
Option	$1\sqrt{2}$
Option	$\sqrt{5}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Two forces $\vec{P}$ and $\vec{Q}$ are acting on an object. Another force $\vec{R}$ is also acting on this object (this force is not parallel to the plane containing $\vec{P}\,\&\,\vec{Q}$). Then resultant $\vec{P}+\vec{Q}+\vec{R}$ of these three vectors
Type	multiple choice
Option	Canf be zero
Option	Cannot be zero
Option	Lies in the plane of $\vec{P}+\vec{Q}$
Option	Lines in the plane of $\vec{P}\,\&\,\vec{P}+\vec{Q}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Let $\vec{C}=\vec{A}+\vec{B}$ then
Type	multiple choice
Option	$\lvert\vec{C}\rvert$ is always greater than $\lvert\vec{A}\rvert$
Option	It is possible to have $\lvert\vec{C}\rvert<\lvert\vec{A}\rvert$ and $\lvert\vec{C}\rvert<\lvert\vec{B}\rvert$
Option	C is always equal to $A+B$
Option	C is never equal to $A+B$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In vector diagram shown in figure where $(\vec{R})$ is the resultant of vectors $(\vec{A})$ and (B).  If $R=\frac{B}{\sqrt{2}}$, then value of angle θ is:
Type	multiple choice

Option	30^0
Option	45^0
Option	60^0
Option	75^0
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The vector that must be added to the vector $\hat{i}-3\hat{j}+2\hat{k}$ and $3\hat{i}+6\hat{j}-7\hat{k}$ so that the resultant vector is a unit vector along the y-axis is:
Type	multiple choice
Option	$4\hat{i}+2\hat{j}+5\hat{k}$
Option	$-4\hat{i}-2\hat{j}+5\hat{k}$
Option	$-3\hat{i}+4\hat{j}+5\hat{k}$
Option	Null vector
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Following sets of three forces act on a body. Which of the following cannot have zero resultant?
Type	multiple choice
Option	10, 10, 10
Option	10, 10, 20
Option	10, 20, 30
Option	10, 20, 40
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	If three forces $\vec{F}_1=3i-4\hat{j}+5\hat{k}$, $\vec{F}_2=-3\hat{i}+4\hat{j}$ and $\vec{F}_3=5\hat{k}$ are acted on a body, then the direction of resultant force on the body is
Type	multiple choice
Option	Along x - axis
Option	Along y - axis
Option	Along z - axis
Option	In indeterminate form
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Two forces, each of magnitude F have a resultant of the same magnitude F. The angle between the two forces is
Type	multiple choice
Option	45^0
Option	120^0
Option	50^0
Option	60^0
Answer	2

Solution	
Positive Marks	4
Negative Marks	1
Question	If vectors, P, Q and R have magnitude 5, 12 and 13 units and $\vec{P}+\vec{Q}=\vec{R}$, then angle between Q and R is
Type	multiple choice
Option	$\cos^{-1}\frac{5}{12}$
Option	$\cos^{-1}\frac{5}{13}$
Option	$\cos^{-1}\frac{12}{13}$
Option	$\cos^{-1}\frac{7}{13}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	There are two force vectors, one of 5 N and other of 12 N, at what angle the two vectors be added to get resultant vector of 17 N, 7N and 13 N respectively
Type	multiple choice
Option	0°, 180° and 90°
Option	0°, 90° and 180°
Option	0°, 90° and 90°
Option	180°, 0° and 90°
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Two vectors $\vec{A}$ and $\vec{B}$ are such that $\vec{A}+\vec{B}=\vec{C}$ and $A^2+B^2=C^2$. Which of the following statement is correct?
Type	multiple choice
Option	$\vec{A}$ is parallel to $\vec{B}$
Option	$\vec{A}$ is anti-parallel to $\vec{B}$
Option	$\vec{A}$ is perpendicular to $\vec{B}$
Option	$\vec{A}$ and $\vec{B}$ are equal to magnitude
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Two forces 3N and 2N are at an angle θ such that the resultant is R. The first force is now increased to 6N and the resultant become 2R. The value of θ is
Type	multiple choice
Option	30°
Option	60°
Option	90°
Option	120°
Answer	4
Solution	
Positive Marks	4

Negative Marks	1
Question	The maximum and minimum magnitude of the resultant of two given vectors are 17 units and 7 unit respectively. If these two vectors are at right angles to each other, the magnitude of their resultant is
Type	multiple choice
Option	14
Option	16
Option	18
Option	13
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The 9. x - component of a force of 50 N is 30 N, then what will be the y – component of the same applied force?
Type	multiple choice
Option	20 N
Option	30 N
Option	40 N
Option	50 N
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A force is inclined at 60° to the horizontal. If its rectangular component in horizontal direction is 50 N. The magnitude of the force and its vertical component respectively are.
Type	multiple choice
Option	100 N, 50 N
Option	$50\sqrt{3}$ N, 100 N
Option	100 N, $50\sqrt{3}$ N
Option	100 N, 100 N
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If a unit vector ĵ is rotated through an angle of 45° anticlockwise, then the new vector will be:
Type	multiple choice
Option	$\sqrt{2}\hat{i}+\sqrt{2}\hat{j}$
Option	$\hat{i}+\hat{j}$
Option	$\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}$
Option	$-\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{j}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A vector lying in x-y plane has a magnitude 3, and makes an angle 30° with the

	x-axis. Find its components along X and Y axis respectively.
Type	multiple choice
Option	$\frac{3}{2},\frac{\sqrt{3}}{2}$
Option	$\frac{3}{2},\frac{3\sqrt{3}}{2}$
Option	$\frac{\sqrt{3}}{2},\frac{3}{2}$
Option	$\frac{3\sqrt{3}}{2},\frac{3}{2}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The angles which a vector $\hat{i}+\hat{j}+\sqrt{2}\hat{k}$ make with X, Y and Z axes respectively are
Type	multiple choice
Option	60°, 60°60°
Option	45°,45°45°
Option	60°, 60°45°
Option	45°, 45°60°
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The angle between the vectors $\left(\hat{i}+\hat{j}\right)$ and $\left(\hat{j}+\hat{k}\right)$ is
Type	multiple choice
Option	30^0
Option	45^0
Option	60^0
Option	90^0
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Consider two vectors $\vec{F}_1=2\hat{i}+5\hat{k}$ and $\vec{F}_2=3\hat{j}+4\hat{k}$. The magnitude of the scalar product of these vectors is
Type	multiple choice
Option	20
Option	23
Option	$5\sqrt{33}$
Option	26
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	If a vector $2\hat{i}+3\hat{j}+8\hat{k}$ is perpendicular to the vector $4\hat{j}-4\hat{i}+\alpha\hat{k}$. Then the value of α is
Type	multiple choice

Option	-1
Option	$\frac{1}{2}$
Option	$-\frac{1}{2}$
Option	1
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Find $\lvert\vec{a}\rvert$ and $\lvert\vec{b}\rvert$ if $\left(\vec{a}+\vec{b}\right)\cdot\left(\vec{a}-\vec{b}\right)=8$ and $\lvert\vec{a}\rvert=8\lvert\vec{b}\rvert$.
Type	multiple choice
Option	$a=\frac{32\sqrt{2}}{3\sqrt{7}}, b=\frac{4\sqrt{2}}{3\sqrt{7}}$
Option	$a=\frac{16\sqrt{2}}{3\sqrt{7}}, b=\frac{2\sqrt{2}}{3\sqrt{7}}$
Option	$a=\frac{8\sqrt{2}}{\sqrt{7}}, b=\frac{\sqrt{2}}{\sqrt{7}}$
Option	$a=\frac{8\sqrt{2}}{\sqrt{7}}, b=\frac{2\sqrt{2}}{\sqrt{7}}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	If for two vectors $\vec{A}$ and $\vec{B}$, sum $\left(\vec{A}+\vec{B}\right)$ is perpendicular to the difference $\left(\vec{A}-\vec{B}\right)$. The ratio of their magnitude is
Type	multiple choice
Option	1
Option	2
Option	3
Option	None of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	If $\hat{n}=a\hat{i}+b\hat{j}$ is perpendicular to the vector $\left(\hat{i}+\hat{j}\right)$, Then the value of a and b may be:
Type	multiple choice
Option	1, 0
Option	$-2, 0$
Option	$3, 0$
Option	$\frac{1}{\sqrt{2}}-\frac{1}{\sqrt{2}}$
Answer	4
Solution	
Positive Marks	4

Negative Marks	1
Question	Find the projection of the vector $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on the vector $\hat{b} = \hat{i} + 2\hat{j} + \hat{k}$?
Type	multiple choice
Option	$\frac{2\sqrt{6}}{3}$
Option	$\frac{4\sqrt{6}}{3}$
Option	$\frac{5\sqrt{6}}{3}$
Option	None of these
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A force $\vec{F} = 5\hat{i} + 6\hat{j} + 4\hat{k}$ acting on a body, produces a displacement $\vec{S} = 6\hat{i} - 5\hat{k}$ Work done by the force is (Hint : $W = \vec{F}.\vec{S}$)
Type	multiple choice
Option	10 units
Option	18 units
Option	11 units
Option	5 units
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle moves from position $3\hat{i} + 2\hat{j} - 6\hat{k}$ to $14\hat{i} + 13\hat{j} + 9\hat{k}$ due to a uniform force of $(4\hat{i} + \hat{j} + 3\hat{k})$N. If the displacement is in meters, then work done will be (Hint: $W = \vec{F} \cdot \vec{S}$)
Type	multiple choice
Option	100 J
Option	200 J
Option	300 J
Option	250 J
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle moves with a velocity $6\hat{i} - 4\hat{j} + 3\hat{k}\,m/s$ under the influence of a constant force $\vec{F} = 20\hat{i} + 15\hat{j} - 5\hat{k}$N. The instantaneous power applied to the particle is (Hint: $P = \vec{F} \cdot \vec{v}$)
Type	multiple choice
Option	35 J/s
Option	45 J/s
Option	25 J/s
Option	195 J/s
Answer	2
Solution	
Positive Marks	4

Negative Marks	1